📊 DESCRIPTIVE AND INFERENTIAL STATISTICS ASSIGNMENT

Problem Statement:
This assignment aims to develop foundational understanding in Descriptive and Inferential Statistics, covering essential concepts such as Measures of Central Tendency (Mean, Median, Mode), Measures of Dispersion (Range, Variance, Standard Deviation), and key inferential methods like Confidence Intervals and Hypothesis Testing. By completing this assignment, students will gain theoretical knowledge and problem-solving skills necessary for data interpretation and statistical analysis.

Q1. Key Statistical Definitions

Objective: Understand foundational statistical terms.
Problem: Write short definitions (2-3 lines each) for the following:
a) Population and Sample
b) Descriptive Statistics and Inferential Statistics
c) Parameter and Statistic
d) Qualitative and Quantitative Data

Q2. Measures of Central Tendency - Definitions

Objective: Learn basic concepts of data centering.
Problem: Define the following terms with one example each:
a) Mean
b) Median
c) Mode

Q3. Manual Calculation of Mean, Median, and Mode

Objective: Apply manual formulas to real data.
Problem:
Given the dataset:
12,18,14,16,18,20,18,1512, 18, 14, 16, 18, 20, 18, 15
Calculate:
a) Mean
b) Median
c) Mode

Q4. Levels of Measurement

Objective: Understand classification of data types.
Problem: Define and give one example for each level of measurement:
a) Nominal
b) Ordinal
c) Interval
d) Ratio

Q5. Variance and Standard Deviation - Theory

Objective: Understand spread/variability in data.
Problem:
a) Define Variance and Standard Deviation.
b) Explain why Standard Deviation is more interpretable than Variance.

Q6. Manual Calculation - Variance and Standard Deviation

Objective: Practice computing data spread.
Problem:
Given the data:
8,10,12,14,168, 10, 12, 14, 16
Calculate:
a) Sample Variance
b) Sample Standard Deviation

Q7. Range and Interquartile Range (IQR)

Objective: Use position-based dispersion metrics.
Problem:
Given the dataset:
22,29,25,31,35,40,45,48,5022, 29, 25, 31, 35, 40, 45, 48, 50
a) Arrange data in ascending order
b) Calculate the **Range**
c) Find **Q1**, **Q3**, and **IQR**

Q8. Five-number Summary and Boxplot Concept

Objective: Summarize distribution of data.
Problem:
Define the Five-number Summary and explain each component:

- Minimum

- Q1 (First Quartile)
- Median
- Q3 (Third Quartile)
- Maximum

  Also, describe how boxplots help in detecting outliers.

Q9. Confidence Interval for the Mean

Objective: Estimate population mean using sample data.
Problem:
A sample of 36 students has an average height of 162 cm with a standard deviation of 6 cm.
Calculate the 95% Confidence Interval for the population mean.
(Hint: Use Z = 1.96 for 95% confidence)

Q10. Hypothesis Testing - One Sample Z-Test

Objective: Make decisions using statistical testing.
Problem:
The average salary in a city is ₹30,000. A random sample of 49 employees has an average salary of ₹31,000 with a standard deviation of ₹4,900.
Test the hypothesis at 5% level of significance to determine if the average salary has increased.
a) State the null and alternative hypothesis
b) Calculate the Z-score
c) Conclude the result using critical value (±1.96)